NaraWear — Core Web Vitals & Performance Optimization Report

Target: **narawear.com** | Host: **Azure App Service (Express.js) + Vite/React SPA** | Date: **November 2, 2026** | Status: **Production Ready**

#	Area / Priority	Action Item	Implementation Details & Technical Solution	Status
1	P0 (Critical)	**Stop 12.7 MB UGC Video Drain**	Added preload="none" and poster images to all UGC video elements. Replaced offscreen autoplay with a lazy IntersectionObserver that only streams/plays a video when scrolled into view and pauses it when scrolled away.	DONE
2	P0 (Critical)	**Eliminate 3.5s Hero Image Delay**	Added <link rel="preload" as="image" fetchpriority="high"> in index.html for mobile (800w) and desktop (1200w). Also configured Azure Express server (server/index.js) to emit early HTTP Link preload headers on /.	DONE
3	P0 (Critical)	**Combine & Defer Tracking Scripts**	Removed duplicate hardcoded gtag.js (AW-17830057970). Initialized dataLayer/fbq stubs early, but deferred GTM container and Meta Pixel network execution until after window.onload, freeing ~590 KB (~330 KB unused) from main-thread CPU.	DONE
4	P0 (Critical)	**Fix Render-Blocking CSS**	Enabled cssCodeSplit: true in Vite. Removed broken font preload tag in index.html that caused 404 console errors. All @font-face declarations now enforce font-display: swap.	DONE
5	P1 (Images)	**Right-Size Collection Images**	Updated imageOptimizer.js with default quality=70 & format=webp. Implemented responsive srcset (360w, 540w, 720w) and tailored sizes across product cards, saving ~240 KB per view.	DONE
6	P1 (CLS)	**Add Width & Height to Images**	Added explicit width and height attributes to hero banners, mystery banner, bento grid tiles, and carousel cards to prevent layout shifts (CLS already down to 0.009).	DONE
7	P1 (JS Bundle)	**Code-Splitting & Modern Target**	Configured Vite build target to modern es2022 with manual vendor chunks (react-vendor, router-vendor, redux-vendor, ui-vendor). All page routes lazy-loaded via React.lazy.	DONE
8	P1 (Caching)	**Long-Term Asset Caching**	Configured Azure App Service Express static middleware to serve hashed JS, CSS, fonts, and images with Cache-Control: public, max-age=31536000, immutable.	DONE
9	P2 (Arch)	**Pre-rendering / Server-side Rendering**	Long-term architectural roadmap item (Next.js / Hydrogen / SSG). Prerender.io middleware currently active in Express to deliver pre-rendered snapshots for search engine and AI crawlers.	PLANNED
10	Accessibility	**Add aria-label to Icon Buttons**	Added accessible aria-label attributes to Cart button, Mobile Menu toggle, Close button, Theme switch, Profile button, Scroll-to-Top, Video Play/Pause, and UGC voting buttons.	DONE
11	Accessibility	**Fix Low-Contrast Text**	Refined text contrast, darkened muted labels, and heightened readability overlays across product cards and bento badges.	DONE
12	Accessibility	**Touch Targets (min 24x24 px)**	Expanded interactive touch hitboxes across header navigation icons, close buttons, and swatches to a minimum of 36x36 px with proper padding and tap spacing.	DONE
13	Accessibility	**Wrap Page Content in <main> Tag**	Wrapped the top-level application routing <Outlet /> inside semantic <main id="main-content" role="main"> in App.jsx.	DONE
14	Best Practices	**Clear Console Errors & Warnings**	Fixed 404 font preload path in index.html, resolved tag mismatched closings, and cleaned up unused tracking tags.	DONE
15	Agentic AI	**Format Standard llms.txt File**	Generated standard-compliant public/llms.txt documenting brand mission, product collections, sizing tools, and policy links for AI search and crawler engines.	DONE

Projected Impact: Mobile Page Weight reduced from **~15.3 MB to ~2.0 MB (-85%)** | Lab LCP projected from **14.6s down to ≤3.8s** | CLS verified at **0.009 (Passing ≤0.1)** | Expected Lighthouse Score: **75–85+**